

## SEMESTER-VII

| UEC719: 5G WIRELESS COMMUNICATION SYSTEM                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                |   |   |       |
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| <p><b>Course Objective:</b> To impart knowledge about wireless communication systems and related design parameters to undergraduate students. To inculcate ability in students to design wireless communication systems, which can provide high data rate to a large number of users. The main goal is to explore the fundamentals of wireless communication systems, including the evolution from 2G to 5G, to understand the principles and technologies behind 5G networks and standards.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                |   |   |       |
| <p><b>Syllabus</b></p> <p><b>Introduction to Wireless and Mobile Communication:</b></p> <p>Evolution of wireless Communication Standards from 2G to 5G, Cellular architecture and fundamental of 2G, 3G, and 4G, Introduction to 5G Usage scenarios, Spectrum for 5G, Spectrum access modes and sharing scenarios, carrier aggregation.</p> <p><b>Wireless Communication Channel Models:</b></p> <p>Free Space Propagation Model, Large-Scale Path Loss model, Overview of outdoor and Indoor propagation models, Small-Scale Fading and Multipath Propagation, fading channel Parameters, Rayleigh, Rician and Nakagami-m Fading Channel Model, 5G Channel Model</p> <p><b>Physical Layer:</b></p> <p>Introduction to diversity and combining techniques, Introduction to MIMO, Alamouti scheme, and channel capacity. Waveform in 4G: OFDM, (transmitter and receiver structure at baseband level), numerology of OFDM in LTE and IEEE 802.11, Waveform in 5G, 5G New Radio (NR), Numerology in 5G waveform, Frame structure in 5G NR, Frequency Ranges, Bandwidths, and Bands for 5G NR, Channel Bandwidth versus UE Channel Bandwidth, Overview of waveform beyond 5G, NOMA.</p> <p><b>5G Network Architecture:</b> Core network architecture, RAN architecture, Main Functional Entities of the 5G Core, High-Level Features of 5G Core, standalone and non-standalone mode, Service data adaptation protocol (SDAP), open RAN, multi-access edge computing (MEC), network function virtualization, Network Slicing, QoS Interworking with Non-3GPP Access Technologies</p> <p><b>Familiarization with standards:</b> IEEE 802.11 to 802.11ac, 802.15, LTE:3GPP TS 21.201, 3GPP TS 36, 3Gpp standard for 5G: 5G NR,</p> |                                                                                                                                |   |   |       |
| <p><b>Laboratory Work:</b> 5G lab is funded by Ministry of Communication and Department of Telecom, India.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                |   |   |       |
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| Experiment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Brief Description                                                                                                              |   |   | Hours |
| 1.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Introduction to 4G VoLTE Smart Phone Section observe the Transmitted/Received RF signals and (I & Q) constellations. Study and |   |   | 2     |

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|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
|    | measure Battery voltages the Battery charging, power management unit in 4G LTE Smart Phone and SIM Interface Section.                                                                                               |   |
| 2. | Study of 4G ATCommands                                                                                                                                                                                              | 2 |
| 3. | To find the bite error rate of EGC (Equal Gain Combining) diversity technique in Rayleigh fading channels. To find the bite error rate of SC (Selection Combining) diversity technique in Rayleigh fading channels. | 2 |
| 4. | To find the bite error rate of MIMO System (Alamouti's Code) in Rayleigh fading channels.                                                                                                                           | 2 |
| 5. | To find the bite error rate of OFDM (Orthogonal Frequency Division Multiplexing ) technique                                                                                                                         | 2 |
| 6. | Demonstration of working of 5G RAN and CORE                                                                                                                                                                         | 4 |
| 7. | Demonstration of concept of network slicing in 5G architecture                                                                                                                                                      | 4 |

### Course Learning Objectives (CLO)

The student will be able to:

1. Understand the fundamental principles of different generations of mobile communication systems.
2. Explore key aspects of 5G New Radio (NR), including waveform numerology, frame structure, and spectrum allocation.
3. Analyse the 5G NR layer using techniques such as OFDM, diversity, and MIMO.
4. Evaluate and analyze various fading channels based on their parameters across different scenarios.
5. Understand the high-level architecture of 5G technology.

### Text Books:

1. *Rappaport, T.S., Wireless Communication-Principles and Practice, Pearson, (2000) 2nd Edition.*
2. *Haykin S & Moher M., Modern Wireless Communication, Pearson, (2005) 3rd Edition.*
3. *Goldsmith, Andrea. Wireless communications. Cambridge university press, 2005.*

### Reference Books:

1. *Lee, William C. Y., Mobile Communication Design and Fundamentals, (1999) 4th Edition.*
2. *Pandya, R., Mobile and Personal Communication System, PHI (2002) 5th Edition.*
3. *Fundamentals of 5g Communications: Connectivity for Enhanced Mobile Broadband and Beyond, Wanshi Chen, Peter Gaal, Juan Montojo, Haris Zisimopoulos , McGrawHill, 2021*
4. *5G Mobile Communications, Wei Xiang, Kan Zheng, Xuemin (Sherman) Shen, Springer International Publishing, January 2017*
5. *IEEE Journal on Selected Areas in Communications IEEE Communications Magazine*